Jian Feng

j36feng@uwaterloo.ca | 647-766-0178 | Website | linkedin/in/fenjian | github/faketut

EDUCATION

University of Waterloo, Computing and Financial Management, Honours, Co-op 09/2023 - 04/2028
Majors: Computer Science, Finance, Statistics (Joint Honours)

EXPERIENCE

Manulife 05/2026 – 08/2026
Co-op
- Resolved global Enterprise Technology Services tickets for cloud and network support.

Treasury Board of Canada Secretariat 09/2025 – 12/2025
Co-op
- Built a .NET MVC module to normalize job titles across 5K public firms via automated pattern analysis.
- Led migration from legacy Selenium to Katalon; set up CI/CD in Azure DevOps to streamline delivery.
- Refactored core parts of an enterprise data platform for maintainability; cut page load 30% via leaner data access.

Lansi Institute of Artificial Intelligence in Medicine 04/2024 – 12/2024
Co-op (Remote)
- Built genomic ETL pipelines for multi-omics, cut preprocessing time 92% via parallel, optimized workflow.
- Led meta-analysis on T2D drug targets, built models linking variants to drug response.

Sanyang Technology Co., Ltd. 07/2024 – 11/2024
Intern
- Built a .NET auto-reconciliation tool that ended manual discrepancy tracking, hit 100% match rate, sped up month-end close.
- Improved ingestion via REST, Kafka, and Redis, handling high-volume sales for near–real-time revenue visibility..

PROJECTS

VoxFlow
- Real-time Twilio Voice agent: bidirectional WS audio into Ultravox; explicit FSM (greet, verify, schedule) for coherent sub-second streaming turns.
- n8n post-call/scheduling: calendar APIs, TZ-aware slots, FastAPI webhooks—bookings/follow-ups without manual ops.
- FastAPI split: thin Twilio HTTP routes, async WS media handlers, injectable session/calendar services—business logic off the hot audio path for easier tests.

ArmyOfSafeguards
- Built a 3-layer safety system with rule-based filters, binary classifiers, and a meta-classifier to combine results for real-time checks.
- Fine-tuned DeBERTa-v3 using LoRA; used neutral-context data to lower false alarms in medical and educational topics.
- Built an expert runner to handle parallel tasks and an aggregator to merge model scores, improving detection stability.

GhostPilot
- Windows interview copilot: invisible overlay streams ASR to a text LLM.
- Low-latency audio: WASAPI loopback, partial/final Azure ASR + segmenter; watchdog auto-restart.
- Alt+P vision path: Qt area capture + JPEG → vision LLM; local RAG (`knowledge/`) folded into text-LLM prompts.

CabStream ETL
- Airflow DAGs: NYC TLC taxi → GCS → BigQuery in million-row batches for fare, duration, and route analytics.
- dbt curated layers (staging → marts, freshness, schema tests) keeping driver/trip/revenue metrics stable across refreshes.
- Looker Studio on BigQuery—filterable KPIs for zone/time revenue and utilization vs. ad hoc spreadsheets.

wlp4-compiler
- End-to-end WLP4 compiler (C++): hand-written lexer, SLR(1) parser, semantic passes on the parse tree, stack-aware ARM64 codegen for the course subset.
- Scoped symbol tables + types across decls/exprs/calls; lowered control flow, procedures, pointers/arrays to ARM64 with stable calling convention and stack layout.

MiniKV
- Minimal LSM: append-only WAL, memtables flushed to SSTables, compaction merging overlaps to cap read amplification.
- SSTable block indexes; WAL replay + partial-tail discard so kill -9 mid-write does not corrupt committed keys.
- Tuned flush/compaction triggers and read-path buffer reuse for higher throughput under concurrent writers/scans vs. naive flush-every-write.

Certificates & Skills

Coursera: AWS Cloud Solutions, CAPM, PMI-RMP, SAS Programmer, Duke Data Engineering, IBM AI for Cybersecurity, IBM Vector Database, SAP Technology Consultant, Stanford Machine Learning, UCI Database Design & Operational BI, UCI Mathematics for ML, UA Reinforcement Learning, Rice Parallel, Concurrent, and Distributed Programming
Skills: Python, SQL, C#, C++, Java, Git, Go, Kotlin, JavaScript, Linux, Windows, MacOS, .Net, Spring Boot, React, Node.js, Express.js, Flask, PostgreSQL, MySQL, MongoDB, Redis, AWS, GCP, Azure, Docker, Kubernetes, Terraform, GitHub Actions, Jira, Confluence, Pytorch, TensorFlow, Scikit-learn, OpenCV, GenAI tools